

HARRY SOOD

San Diego, CA | (901) 921-1040 | harry.hs.sood@gmail.com | US Citizen | Active DoD Secret Clearance

SKILLS

Programming	Python (Pandas, NumPy, Matplotlib, SciPy, OpenPyXL) SQL JavaScript MATLAB C++
Cloud & Infrastructure	Azure (Functions, Durable Functions, ADLS, Blob Storage) GCP (Cloud Storage, Google Sheets API)
Data & Analytics	Excel Jupyter Notebooks Power BI ETL Pipeline Development Statistical Analysis Signal Processing
Tools & Software	Visual Studio Code Git Windows, macOS, Linux Microsoft Office Suite Agile Workflow

EXPERIENCE

Senior Data Engineer

July 2022 – Present

General Atomics Aeronautical Systems

San Diego, CA

Generative AI Data Pipeline Development

- Architected and implemented a scalable Azure data pipeline to ingest, classify, and route proprietary company documents from OnBase into Azure ADLS blobs, enabling production-ready AI-driven indexing
- Deployed durable Azure Functions with eternal orchestration to automatically process 50,000+ documents on initial load and an average of 50 additional documents per week; eliminated IT-dependent manual processes
- Engineered fault tolerance via file-by-file orchestration, Azure Table state tracking, batch finalization to resolve race conditions, and scheduled data validation; achieved zero data loss across all environments
- Hardened pipeline with environment-driven configurability, Git-based version control, and local Linux testing to accommodate shifting upstream conventions and evolving requirements without service interruption

Flight Data Analysis

- Conducted 100+ flight log analyses — parsing telemetry data from a repository of 150,000+ logs across 14 UAV variants, each with 1,300+ signals — to support investigations, establish performance baselines, and train junior engineers
- Develop custom event detection scripts and synthetic signals to identify behavioral patterns and anomalies across datasets spanning as many as 200,000 events per analysis
- Deliver tailored visualizations and Power BI dashboards scaled to dataset size and stakeholder needs, enabling engineering and program teams to interpret complex flight data and inform decisions
- Ensure signal integrity across a diverse fleet by processing and filtering downlinked telemetry data using statistical outlier analysis, threshold-based detection, and aeronautical domain expertise

Root Cause Analysis (RCA) & Cross-Functional Consulting

- Operate as an internal consultant on two to three multi-month RCA investigations per year, jumping into unfamiliar subsystems, rapidly becoming a domain expert, and delivering unbiased findings on failures that eluded other teams
- Interface directly with internal engineering teams, program management, and external customers to define analysis requirements, present conclusions, and translate complex data insights into actionable decisions
- Designed a preliminary data pipeline bridging engineering and manufacturing operations for a new aircraft program, enabling the machine shop to track part schedules; helped accelerate first flight to meet a year-long program timeline
- Supplement data analysis with hands-on participation in operations including engine tests, starter generator inspections, and datalink troubleshooting to validate and contextualize results

Co-Founder, Data & Infrastructure

July 2022 – Present

chipsID LLC – IC Counterfeit Mitigation Services

San Diego, CA

Database & Cloud Infrastructure Development

- Established GCP Cloud Storage infrastructure with automated Python backend to manage part image repositories, generate links, and orchestrate file uploads across buckets; enabled platform-agnostic image sourcing and fast retrieval
- Implemented a MySQL database on GCP for component tracking, then migrated to a Google Sheets-based data layer after evaluating cost, team accessibility, and efficiency; leveraged built-in revision control for metadata collaboration
- Built JavaScript backend features including advanced filtering, search persistence, and API integration to connect GCP Cloud Storage and metadata to the web platform for seamless part discovery

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EXPERIENCE (CONTINUED)

Project Engineer

July 2021 – June 2022

General Atomics Aeronautical Systems

San Diego, CA

Software Tools & Data Systems

- Expanded software tools suite from 4 to 9 custom applications, maintaining full ownership across backend logic, CLI and Excel-based interfaces, and comprehensive documentation; ensured team-wide usability through training and support
- Integrated tools across multiple data sources including GOLD depot, SharePoint, Power BI, and legacy systems to track component obsolescence, LRU replacements, and program supply chain risk; delivered insights to external stakeholders
- Structured tools with upstream change resilience; achieved zero runtime logic issues after team deployment; upon request, returned post-departure to enhance and add features based on evolving program needs

Modernization & Obsolescence Mitigation

- Spearheaded a successful \$3M tech refresh of the Certifiable Ground Control Station's (CGCS) computer systems
- Redesigned the CGCS's C-Band video architecture to minimize failure points and non-recurring engineering costs
- Executed mitigation strategies to defend against supply chain shortages and obsolescence of critical components

Engineering Intern

June 2020 – August 2020

General Atomics Aeronautical Systems

San Diego, CA

Software Tools Development

- Created a custom suite of interlinked tools to increase team efficiency, resulting in estimated annual savings of 780 engineer labor hours (\$195,000/year at \$250/hour)
- Programmed data integration software pulling from Windchill, SAP, and team-managed spreadsheets to track the impact of obsolete parts on associated end products

Researcher

May 2018 – April 2020

University of Michigan, College of Engineering

Ann Arbor, MI

Formally Verified Autonomous Systems

- Validated and verified autonomous vehicle systems through formal proofs for Toyota-sponsored research program
- Modeled autonomous vehicle collision avoidance through robust simulations under various dynamic parameters

EDUCATION

M.Eng. Space Engineering | GPA: 4.0/4.0 | Focus: Systems Design

Class of 2021

University of Michigan, College of Engineering

Ann Arbor, MI

B.S.E. Aerospace Engineering | GPA: 3.4/4.0 | Minor: Computer Science

Class of 2020

University of Michigan, College of Engineering

Ann Arbor, MI

PUBLICATIONS

Aakash Abhishek, **Harry Sood**, Jean-Baptiste Jeannin. 2020. Formal verification of swerving maneuvers for car collision avoidance. ACC 2020. IEEE. URL: <https://doi.org/10.23919/ACC45564.2020.9147679>

Aakash Abhishek, **Harry Sood**, Jean-Baptiste Jeannin. 2020. Formal verification of braking while swerving in automobiles. HSCC 2020. ACM. URL: <https://doi.org/10.1145/3365365.3382217>

CERTIFICATIONS

FAA Private Pilot Certificate | Aircraft Single Engine Land (ASEL)